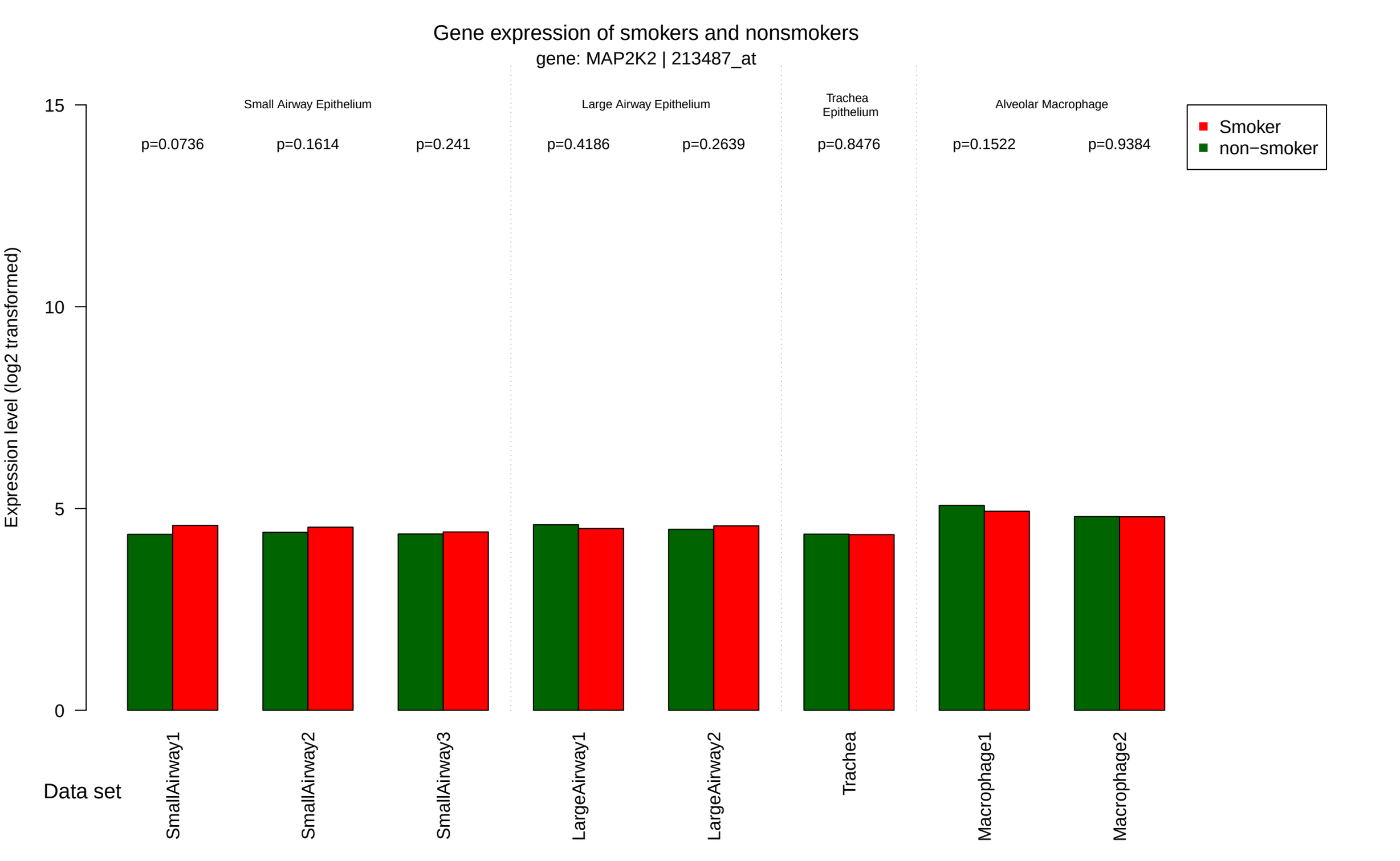
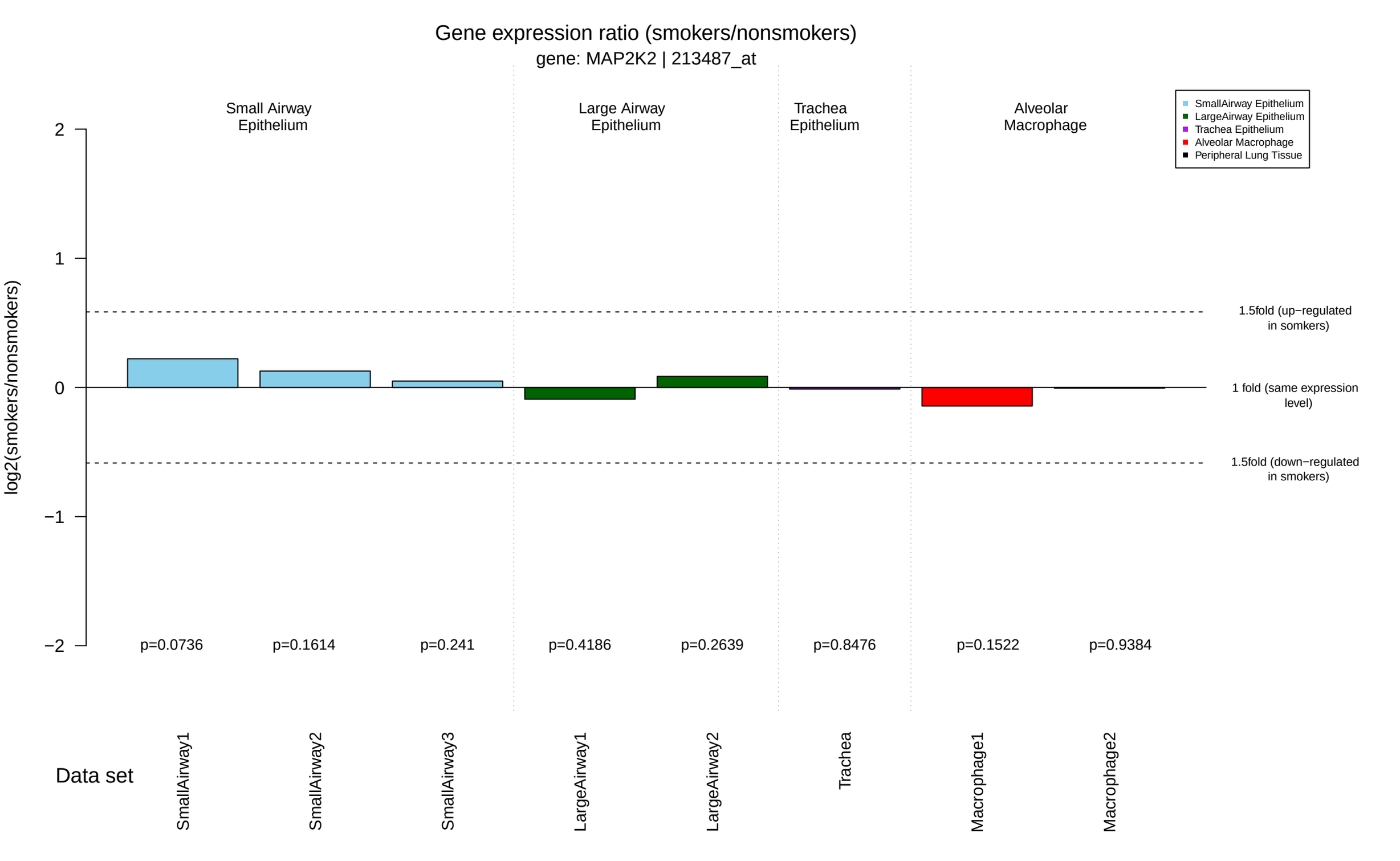
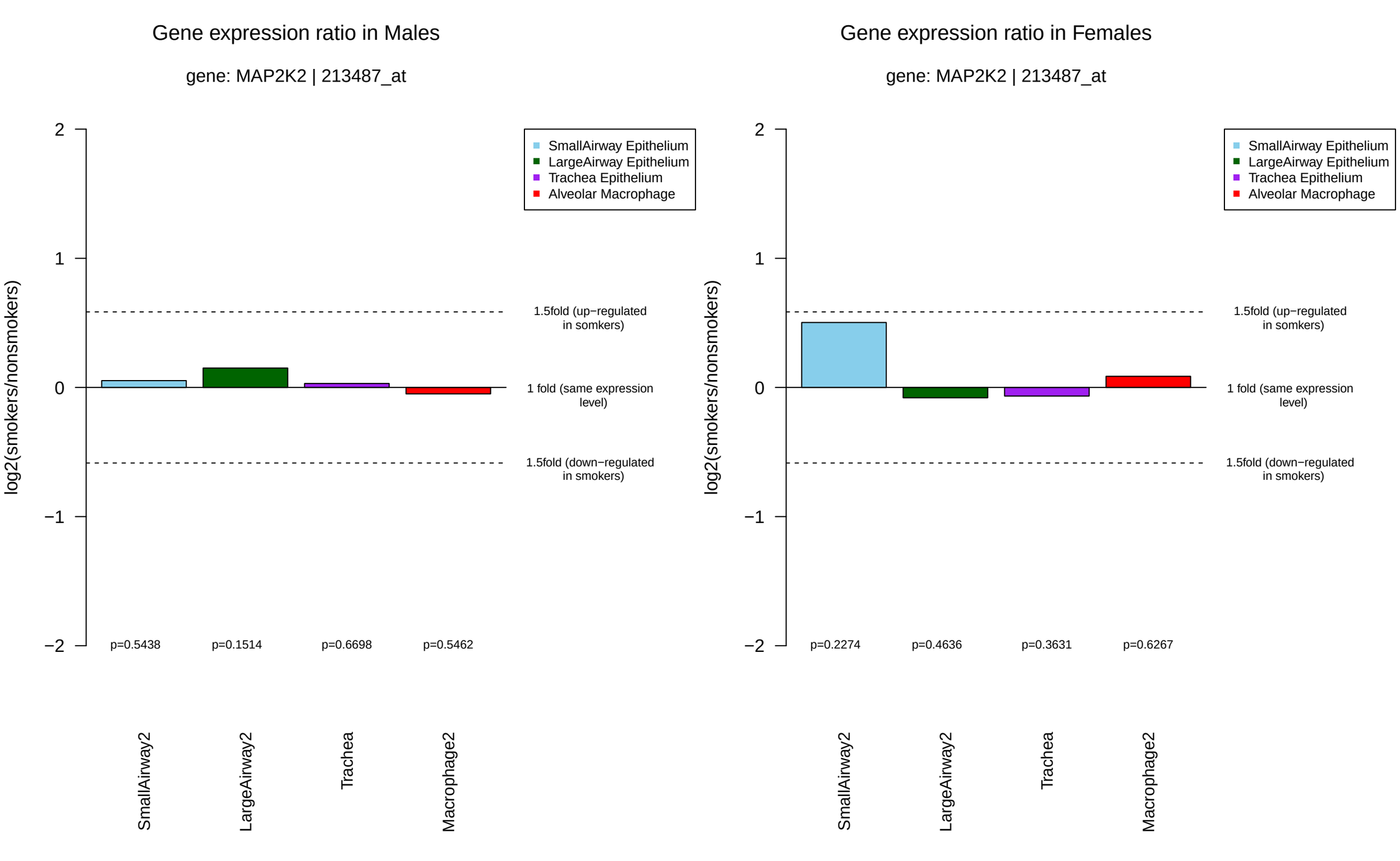
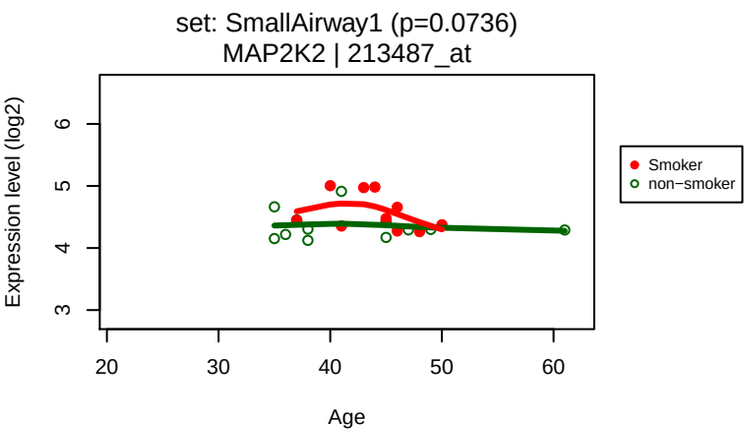
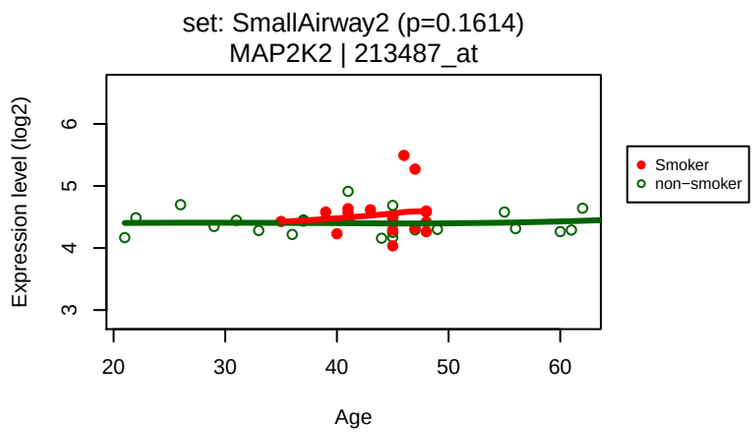
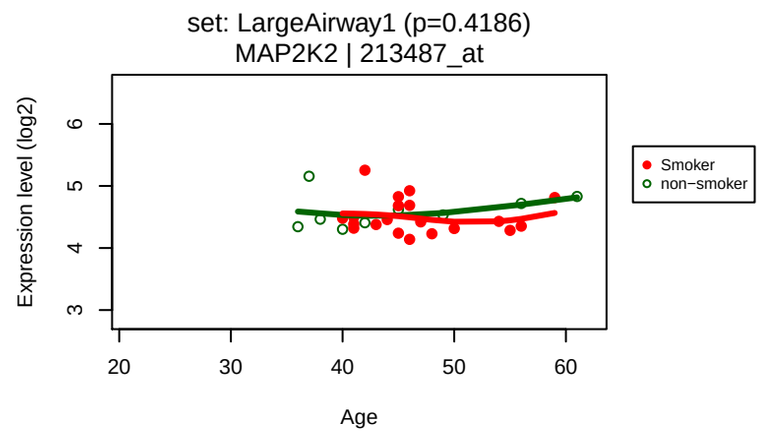

Gene expression of smokers and nonsmokers
gene: MAP2K2 | 213487_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0736
p=0.1614
p=0.241
p=0.4186
p=0.2639
p=0.8476
p=0.1522
p=0.9384
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

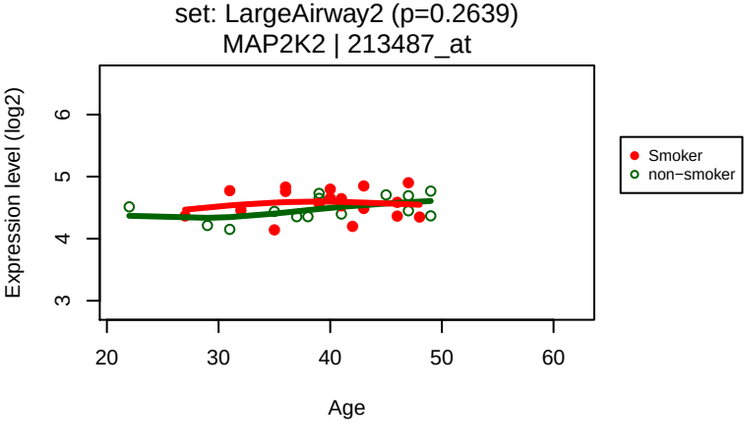

Gene expression ratio (smokers/nonsmokers)
gene: MAP2K2 | 213487_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0736
p=0.1614
p=0.241
p=0.4186
p=0.2639
p=0.8476
p=0.1522
p=0.9384
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

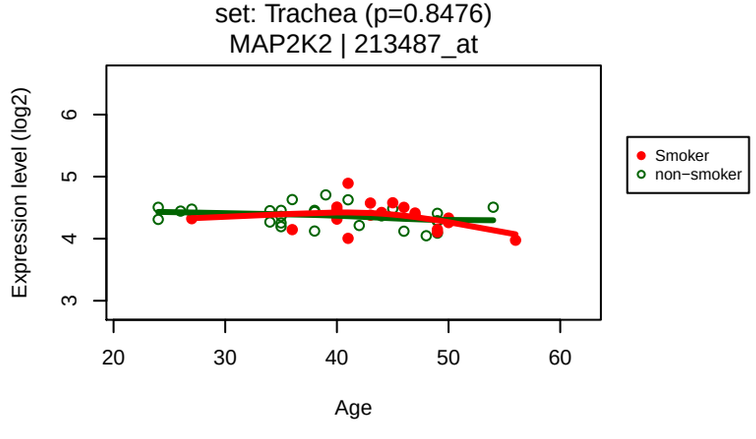

Gene expression ratio in Males
gene: MAP2K2 | 213487_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5438
p=0.1514
p=0.6698
p=0.5462
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAP2K2 | 213487_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2274
p=0.4636
p=0.3631
p=0.6267
SmallAirway2
LargeAirway2
Trachea
Macrophage2

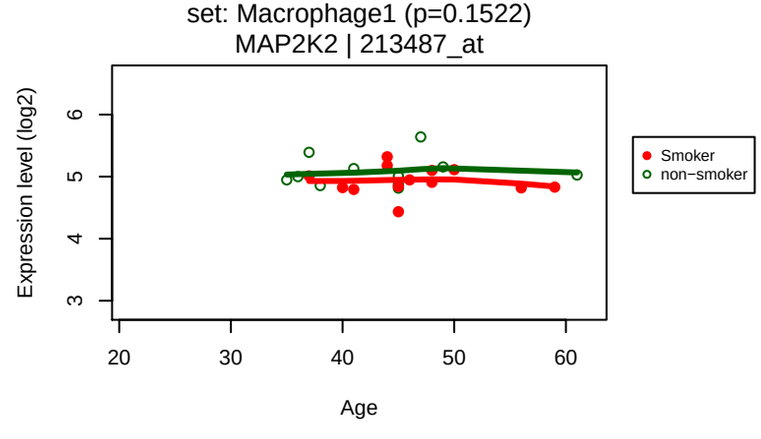

set: SmallAirway1 (p=0.0736)
MAP2K2 | 213487_at
Expression level (log2)
Age
Smoker
non−smoker

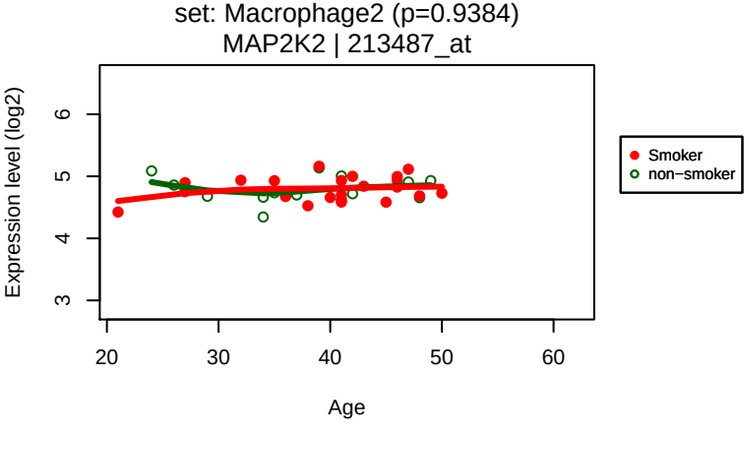

set: SmallAirway2 (p=0.1614)
MAP2K2 | 213487_at
Expression level (log2)
Age
Smoker
non−smoker

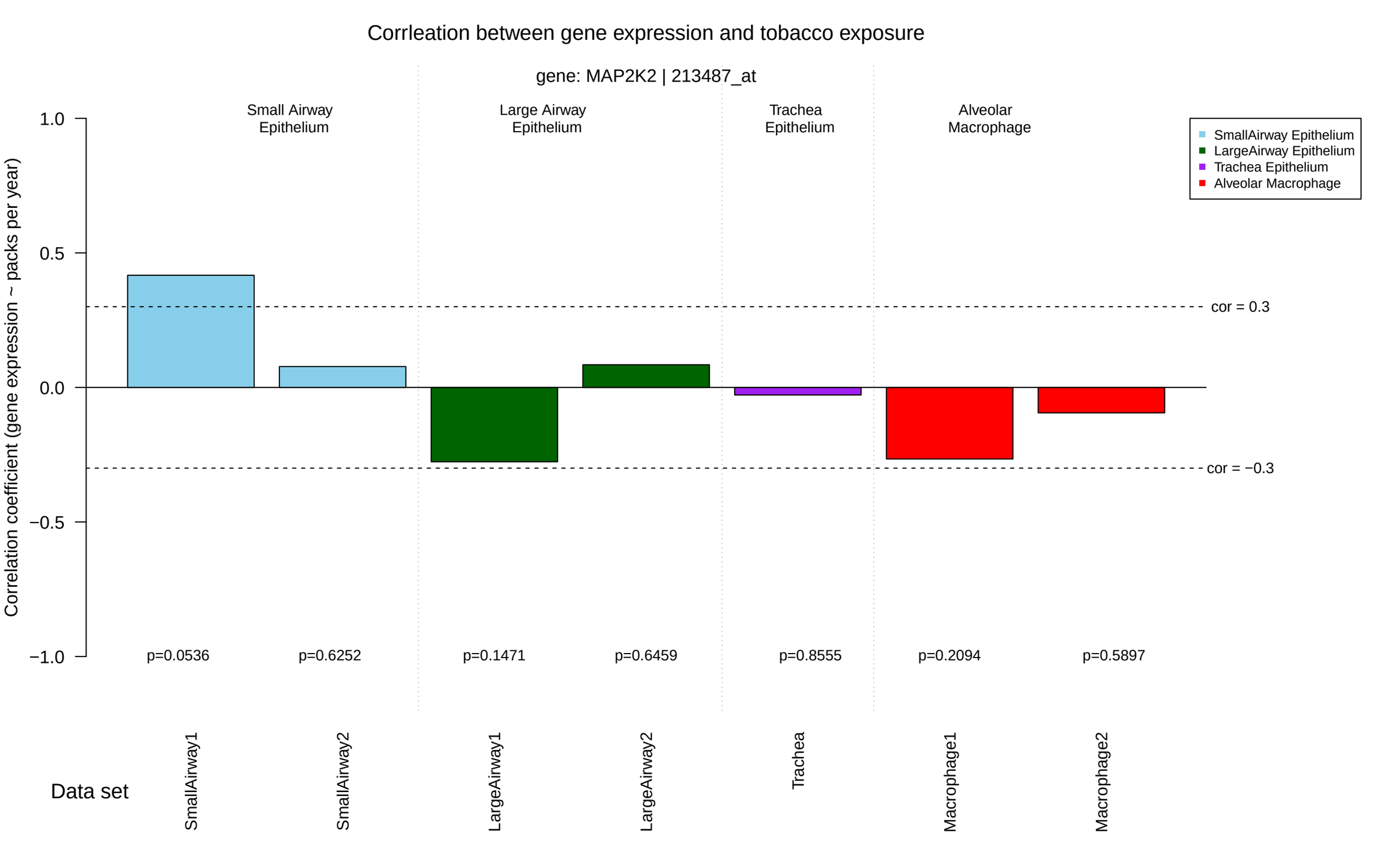

set: LargeAirway1 (p=0.4186)
MAP2K2 | 213487_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2639)
MAP2K2 | 213487_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8476)
MAP2K2 | 213487_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1522)
MAP2K2 | 213487_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9384)
MAP2K2 | 213487_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAP2K2 | 213487_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0536
p=0.6252
p=0.1471
p=0.6459
p=0.8555
p=0.2094
p=0.5897
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2